

Hamza Masri

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Profile

DevOps Engineer and AI Developer with 2.5+ years at DVZ M-V GmbH, where I'm also writing my Master's thesis (M.Sc. Computer Science, University of Rostock) on context provisioning for self-hosted coding agents. Built and shipped multiple products end-to-end alongside my professional role.

Seeking a full-time position as an AI Engineer in Munich, focused on production-ready AI systems. Strong in both practice and theory: hands-on experience in LLM integration, prompt engineering, and AI automation, backed by AI-focused Bachelor's and Master's theses.

Professional Experience

DevOps Engineer & Kubernetes Specialist ————— 11/2024 – Present
DVZ M-V GmbH, GEW Department, Rostock

- Managing containerized microservices and CMS-based applications with Kubernetes.
- Increasing deployment efficiency and reducing manual configuration through automation with Helm and CI/CD.
- Enhancing system stability and minimizing downtime through proactive monitoring with Prometheus and Grafana.
- Development and implementation of a Keycloak Helm Chart for centralized Single Sign-On.
- Technologies: *Python, PHP, YAML, Kubernetes, Docker Desktop*

Bachelor Thesis - Working Student in Machine Learning ————— 07/2024 - 11/2024
DVZ M-V GmbH · University of Rostock - Final Grade: 1.8

- Development of an AI-powered search system for geographic information retrieval and user classification.
- Implementation of vector search and language models to improve search quality and user experience.
- Implementation of an AI classifier to identify user groups (beginner/expert) for personalized search.
- Development of AI-based solutions for geographic data analysis.
- Implementation of machine learning algorithms for data processing.
- Technologies: *PHP, Python, PyTorch, OpenAI*.

Software Development Intern (Full-Stack) ————— 03/2024 – 07/2024
DVZ M-V GmbH · University of Rostock

- Development and maintenance of web applications for geographic information systems (GIS).
- Development with the OpenStreetMap API for processing geographic data.
- Creation and maintenance of *PHP tests* to ensure code quality.

- Use of *jQuery* and *HTML5* for interactive frontend elements.
- Technologies: *PHP*, *JavaScript*, *HTML5*, *CSS*, *SQL*, *OpenStreetMap API*.

Freelance AI Developer & Full-Stack Developer ————— 2021 – 2026
Self-employed · Online

- Development of customized websites and applications for various clients as a Full-Stack Developer.
- Consulting and implementation of individual software solutions for small businesses.
- Integration of modern AI solutions in various business areas (e.g., document analysis, AI chatbots, prompt pipelines, prompt engineering, and more...)
- Technologies: *React*, *PHP*, *Python*, *WordPress*, *JavaScript*, *MySQL*, *OWASP*, *Nginx*, *PyTorch*, *OpenAI*.

Service and Hospitality Employee ————— 2017 – 2024
Various Restaurants, Bars and Hospitality Establishments · Rostock and Jerusalem

- 7 years of experience in the hospitality and service industry, parallel to studies for study financing. Development of communication and multitasking skills as well as successful collaboration in teams under time pressure.
- Responsibility for customer service, billing, and service quality. Flexible adaptation to different working hours and task areas.

Education

Master of Science in Computer Science ————— 2024 – Present
University of Rostock, current GPA: 2.1

- Currently writing Master's thesis in cooperation with DVZ M-V GmbH: *Context Provisioning for Self-Hosted Coding Agents under Resource Constraints: A Controlled Experimental Study* (expected completion: 04/2027).

Bachelor of Science in Computer Science ————— 2020 – 2024
University of Rostock, Final Grade: 1.8

- Degree: Bachelor of Science.
- Focus areas: AI systems, software development, databases, mathematics, algorithms, computer networks, and data security.

Intensive German Course (A1 to B2) ————— 2018 – 2019
Study in Germany – Rostock, Final Grade: 1.3

- Successful completion of an intensive German course from A1 to B2.
- Targeted learning of IT technical terms for better communication in professional contexts.

Shuafat High School, Jerusalem ————— 2012 – 2018
Average Grade: 1.6

- Main subjects: Computer Science, Mathematics, Physics.
- Additional elective courses in computer science to deepen my programming skills.

AI Projects

Mender – Desktop Automation Platform with Self-Healing Selectors (*Own Project*)

- Built an Electron desktop application (React, TypeScript, Node.js/Express, SQLite) that runs scheduled browser automation with Playwright and AI-generated content via the Anthropic Claude API.
- Designed a self-healing selector system: a health check detects broken page selectors, verifies a replacement on the live page, corroborates it across runs, and applies it to the running process without a restart.
- Implemented a campaign scheduler with time windows, pacing, daily limits, and N-way action chains, backed by reusable content pools.
- Built an AI model registry that checks all configured models daily, flags retired ones, and switches calls to a working model automatically.
- Technologies: *TypeScript, Electron, React, Node.js, Express, SQLite, Playwright, Claude API*.

Excessus – AI Client Intake for Law Firms (*Own Venture*)

- Designed and built an AI-powered client intake SaaS for German law firms, based on Retrieval-Augmented Generation (RAG) and the Anthropic Claude API.
- Owned the full product cycle: architecture, implementation, and go-to-market via cold outreach campaigns to law firms.
- Technologies: *Python, Claude API, RAG*.

AI-Powered SEO Content Pipeline (*Freelancing*)

- Development of a full-cycle AI pipeline for automated creation, legal verification, and SEO optimization of specialized texts.
- Use of prompt engineering and streaming (Anthropic Claude API) for generating legally correct outlines and content.
- Implementation of multi-step verification (Outline, Content Generation, Legal Review, SEO Rework, Humanization) to ensure quality and SEO performance.
- Enhancement of prompt engineering skills for professional image creation.
- Development of the frontend for user interaction, including detailed technical documentation for maintenance and further development.

Seal Tracking in Drone Footage with DINOv3 (*University Project, Team of 4*)

- Built a Python pipeline that tracks seals in drone video with almost no labeled data: frame extraction, dense feature extraction with the self-supervised DINOv3 Large vision transformer (no fine-tuning), and PCA visualization separating seals from the water background.
- Compared similarity-based localization (Euclidean distance with neighbourhood aggregation vs. cosine similarity) and integrated the Segment Anything Model (SAM); cosine-guided SAM produced clean, target-specific seal masks where unguided SAM was dominated by background noise.
- Propagated segmentation masks across frames via cosine similarity between local patch neighbourhoods and rendered annotated output videos; evaluated all methods on localization, boundary quality, background noise, and target specificity.
- Technologies: *Python, PyTorch, DINOv3, SAM, NumPy, Jupyter*.

Other Projects

Algorithm Optimization in Competitions (*University Project*)

- Participation in programming competitions for optimizing complex algorithms under time pressure.
- Development of solutions for graph problems, dynamic programming, and search algorithms.
- Technologies: *Python, C#, Java*.

Other Projects: Web applications (React, WordPress, HTML/3), MediaWiki extensions (PHP/JavaScript), responsive design, UI/UX implementations.

Technical Skills

- **AI & Machine Learning:** *PyTorch, OpenAI API, Anthropic Claude API, Prompt Engineering, LLM Integration, Vector Search, scikit-learn, Ollama, LM Studio*
- **Programming Languages:** *C#, Python, PHP, Java*
- **Web Technologies:** *HTML, PHP, React, Angular*
- **Databases:** *MySQL, SQLite3, PostgreSQL*
- **Tools & Systems:** *Git, Docker, Linux, Bash, Windows, Canva, Jira*
- **DevOps & CI/CD:** *Kubernetes, Minikube, Helm, Jenkins, Azure, Prometheus, Grafana*
- **Web Frameworks & APIs:** *Flask, REST APIs, OpenAI, OpenStreetMap API*
- **Automation & Workflows:** *n8n, Python Scripting*
- **Data Processing:** *NumPy, Pandas*
- **Documentation & Design:** *LaTeX, Figma, Adobe, Canva*

Languages

- **German:** (C1) Advanced proficiency
- **English:** (C2) Fluent, near-native proficiency
- **Arabic:** Native speaker
- **Hebrew:** (A2) Basic knowledge